

ORIGINAL ARTICLE

Plasma Proteins Associated With the Immune Response to Anthracycline Cardiotoxicity in Patients With Hematologic Malignancies

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BACKGROUND: Anthracyclines such as doxorubicin are well recognized to induce dose-dependent cardiotoxicity. However, traditional imaging and blood biomarkers of myocardial injury are inadequate for early risk stratification. We aimed to discover plasma proteins that correlate with the subsequent development of anthracycline-induced systolic dysfunction in patients, with validation studies in preclinical models.

METHODS: We used an aptamer-based proteomics platform to measure over 7000 proteins in the plasma obtained at baseline (before anthracycline chemotherapy) in a prospectively enrolled cohort of patients with hematologic malignancies (n=34). The correlation between protein levels and change in left ventricular ejection fraction was assessed. Candidate biomarkers were validated in a second patient cohort treated with anthracyclines for hematologic malignancies (n=32), as well as in a mouse model of delayed doxorubicin cardiotoxicity.

RESULTS: In both discovery and validation cohorts, baseline levels of CPVL (carboxypeptidase vitellogenic-like protein) and PIGR (poly immunoglobulin receptor) measured in the plasma before anthracycline treatment correlated with a subsequent decline in left ventricular ejection fraction (PIGR: $R=-0.74$; $P=0.0022$; CPVL: $R=-0.74$; $P=0.0022$). Findings were confirmed in human plasma samples using ELISA or Western blot, and similar changes in proteins were observed in mice treated with doxorubicin. In further support of PIGR as a biomarker representing the systemic adaptive immune response, doxorubicin treatment was associated with increased circulating levels of IgA⁺ (immunoglobulin A) B cells in mice, which similarly correlated with the decline in left ventricular ejection fraction. The highest quartile of PIGR and CPVL levels was associated with decreased overall survival but not progression-free survival.

CONCLUSIONS: Preanthracycline levels of the plasma immunologic proteins CPVL and PIGR were associated with a subsequent decline in cardiac function, representing potentially new inflammatory biomarkers of anthracycline cardiotoxicity.

Key Words: animals ■ blood protein ■ humans ■ immunoglobulins ■ proteomics

Anthracycline chemotherapies, including doxorubicin, are commonly used in first-line regimens for hematologic malignancies and other cancers.¹ However, anthracyclines are associated with dose-dependent cardiotoxicity in up to 20% of patients treated. This can limit the use of effective cancer treatment and lead to worse cardiac and cancer outcomes, including heart failure

(HF).^{2,3} Although conventional blood biomarkers such as cardiac troponins and pro-BNP (pro-B-type natriuretic peptide) are often elevated after cytotoxic chemotherapy, BNP is not consistently associated with subsequent cardiomyopathy in patients treated with anthracyclines. While cardiac troponins have excellent negative predictive value, their positive predictive value for clinically

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WHAT IS NEW?

- In patients with hematologic malignancies, prechemotherapy circulating levels of PIGR (poly immunoglobulin receptor) and CPVL (carboxypeptidase vitellogenic-like protein) correlate with subsequent systolic dysfunction after anthracycline treatment.
- Mice treated with doxorubicin exhibit increased circulating CPVL, PIGR, and IgA⁺ (immunoglobulin A) B cells, which correlate with systolic dysfunction.
- Elevated PIGR and CPVL in patients are associated with decreased survival but not malignancy progression.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Potential biomarkers of anthracycline-associated cardiotoxicity, such as PIGR and CPVL, may impact clinical practice by enhancing cardiovascular risk stratification in individuals undergoing anthracycline treatment.
- The integration of novel biomarkers of cardiotoxicity, such as CPVL and PIGR, could facilitate early referrals to cardio-oncology specialty care, enable personalized cancer treatment options for patients, and assist with tailored use of cardioprotective therapies.
- These data also build upon our limited knowledge of how the inflammatory response contributes to cardiotoxicity in anthracycline treated patients.

Nonstandard Abbreviations and Acronyms

CPVL	carboxypeptidase vitellogenic-like protein
HF	heart failure
IgA	immunoglobulin A
LVEF	left ventricular ejection fraction
PIGR	poly immunoglobulin receptor
pro-BNP	pro-B-type natriuretic peptide

significant cardiotoxicity is modest.⁴ Therefore, there is an unmet need for new risk stratification methods to identify which patients treated with anthracyclines might develop cardiac dysfunction.

Our group has shown that plasma samples from anthracycline-treated patients can be leveraged for correlative biomarker studies and to guide mechanistic work in preclinical models.^{5,6} As systemic inflammation has been reported in patients treated with anthracyclines, as well as in those with chronic HF, immunologic mediators represent 1 such class of mechanistic biomarkers.^{7–9} Animal models of doxorubicin cardiotoxicity have further revealed elements of both the cardiac innate immune response^{10,11} and a chronic adaptive immune response consisting of B cells and cardiac-directed CD8⁺ T

cells.^{5,12} These studies suggest that doxorubicin induces antigen presentation, leading to cell-mediated and antibody responses; however, the details and relevance of these immune processes are poorly understood. Furthermore, the immune response has yet to be integrated into clinical risk stratification for patients susceptible to doxorubicin cardiotoxicity. We hypothesized that probing the circulating proteome in patients would reveal potential mechanistic biomarkers, including inflammatory markers, for anthracycline cardiotoxicity.

Here we used aptamer-based plasma proteomics in patients with hematologic malignancies to identify biomarker candidates to facilitate early diagnosis of anthracycline-induced decline in left ventricular systolic function. We identified novel immune signatures in humans that were further investigated using mouse models, adding to our understanding of the chronic inflammatory response in anthracycline cardiotoxicity.

METHODS

The data, methods, and materials will be available to other researchers for the purposes of reproducing the results or replicating procedures. Detailed methods and any associated references are provided in the [Supplemental Material](#).

Study Design and Patient Enrollment

In this single-center observational study, adult patients were prospectively enrolled at Beth Israel Deaconess Medical Center from 2019 to 2025 and divided into discovery and validation cohorts. Eligible participants were individuals aged 18 years or older with hematologic malignancies who were scheduled to receive anthracyclines. Participants additionally had >1 risk factor for the development of cardiotoxicity, including >65 years of age, hypertension, treatment for aggressive non-Hodgkin lymphoma, prior chest irradiation, or a planned cumulative doxorubicin dose ≥ 300 mg/m². For patients receiving anthracyclines other than doxorubicin, conversions were made using an equivalent doxorubicin dose based on prior publications.¹³ Patients were excluded if they had a baseline left ventricular ejection fraction (LVEF) <50% or underwent an initial blood draw but did not proceed with the planned chemotherapy. All study subjects provided their consent before enrollment. All study procedures and informed consent forms were approved by the Dana-Farber/Harvard Cancer Center institutional review board (protocol 18-126), with local oversight from the Beth Israel Deaconess Medical Center Cancer Clinical Trials Office.

Participants underwent a baseline transthoracic echocardiogram and blood draw before the initial dose of anthracyclines, and cardiovascular risk factors were assessed via a questionnaire. A follow-up echocardiogram was performed at 12 (± 6) months. LVEF was determined by the Simpson biplane method of discs from apical 2- and 4-chamber views using EchoPAC (Version 206, General Electric) by an experienced cardiologist blinded to patient identification and study time points, with the exception of 10 echocardiograms from other hospitals for which we were unable to access the original images. For these echocardiograms, we relied on outside hospital reports, which

included clinical visual interpretation. Covariates included in the multivariable models were assessed at baseline (before chemotherapy). The Chronic Kidney Disease Epidemiology Collaboration 2021 equation was used for estimated glomerular filtration rate estimation. HF was adjudicated by manual chart review according to published criteria, as our group has previously published.¹⁴ For the purposes of this study, cardiotoxicity was defined as a decrease in LVEF, and for the purposes of immunoblot experiments, we selected patients with a relative LVEF decline of 20% or more. Our initial sample size was based on a prior study from our group showing that the protein hemopexin, as measured by a similar aptamer-based proteomics platform, was associated with a change in cardiac function in patients with breast cancer treated with anthracyclines ($R=-0.59$; $P=5.0\times 10^{-4}$).⁶ The cost of performing comprehensive aptamer-based proteomics was also factored into sample size considerations for the discovery study.

Mouse Doxorubicin Model

All animal studies were approved by the Institutional Animal Care and Use Committees at Tufts University. Mice were maintained under pathogen-free conditions and in a 14-hour light/10-hour dark cycle, at an ambient temperature of 70 to 74 °C, with humidity between 30% and 70%. Male C57BL/6 mice at 8 to 12 weeks of age were dosed with doxorubicin 5 mg/kg or saline by intraperitoneal injection, randomized by cage using a random number generator (<https://www.calculator.net/random-number-generator.html>), weekly for 4 or 8 weeks ($n=6$ /group, 18 total mice). A minimum sample size of 6 mice per group was calculated to detect statistically significant changes in cardiac function with a power of $\geq 80\%$ and $\alpha=0.05$. All analyses were performed in a blinded fashion using an ID number assigned by a third party.

Statistical Analysis

For aptamer-based proteomics, relative fluorescent units were log normalized. In patients, Pearson correlation was used to compare baseline protein levels with the change in Δ LVEF, which was treated as a continuous variable for all analyses. Statistical significance was determined using a Benjamini-Hochberg false discovery rate $q<0.05$. Multivariable adjustment using ordinary least squares linear regression was performed for proteins that met statistical significance based on the false discovery rate q value. Multivariable models using ordinary least squares linear regression were adjusted for clinical characteristics that could potentially affect protein levels, such as age, sex, body mass index, baseline LVEF, and estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation. The top 2 hits were selected for validation and ongoing study based on adjusted P values after multivariable adjustment analysis. For immunoblotting of CPVL (carboxypeptidase vitellogenic-like protein), samples were selected based on a relative LVEF decline of 20% or more. In multivariable-adjusted models, we used a $P<0.05$ to declare statistical significance. Statistical analyses were performed using STATA 14 or Prism 10. In mice, saline and doxorubicin-treated groups were compared using an unpaired t test or 1-way ANOVA with Tukey multiple comparison test. Statistical significance was considered at a 2-tailed $\alpha<0.05$. For cancer outcomes, clinical variables were abstracted from the electronic medical record for each

patient and reviewed by an attending oncologist. Pretreatment biomarkers were compared between groups with disease progression versus nonprogression during the study period and for all-cause mortality during the study period. The Wilcoxon rank-sum test was used to determine significant differences in biomarker levels using a threshold of $P<0.05$. A multivariable Cox proportional hazards model with correction for international prognostic index and age was used to evaluate the association of pretreatment biomarker levels with risk of progression or death. A log-rank test was used to determine the significance of the difference in hazard ratios, both globally for the model and individually for biomarkers, with $P<0.05$ indicating significance. Proteins were modeled continuously with log transformation in each of the Cox proportional hazards models. To verify that the log transformation did not result in a violation of the proportional hazards assumption, the Schoenfeld residuals were tested for each model, each resulting in a $P>0.05$, indicating no violation of the proportional hazards assumption. Kaplan-Meier survival analyses for progression and survival were used to compare strata of pretreatment biomarker levels, with the log-rank test used to test for significant differences among strata with a threshold of $P<0.05$. Statistical analysis for cancer outcomes was performed using R, version 4.4.2.



RESULTS

Immune Markers PIGR and CPVL Correlate With Anthracycline-Associated Cardiac Dysfunction

The discovery cohort consisted of 34 patients with hematologic malignancies enrolled prospectively at our institution from 2018 to 2022. Patients received an average of 276.9 ± 57.4 mg/m² of doxorubicin or equivalent anthracycline. On average, the LVEF declined by 11 ± 14 absolute percentage points, from a mean baseline of $62.1\pm 5.6\%$ to a posttreatment LVEF of $54.9\pm 8.4\%$ over an average of 216.8 ± 68.5 days to follow-up. Although there was some variability in the timing of the follow-up echocardiogram, decreased LVEF did not correlate with the time between serial echocardiograms (Figure S1A). Only 1 patient developed HF. Higher baseline LVEF correlated with subsequently decreased LVEF (Figure S1B). Additional clinical characteristics can be found in Table 1. Using aptamer-based proteomics to measure 7288 proteins, we found that baseline (prechemotherapy) levels of 24 proteins correlated with Δ LVEF using a false discovery rate threshold of $q<0.05$ (Table 2). These included several proteins that participate in immune processes, such as PIGR (poly immunoglobulin receptor), CPVL, and TCPTP (tyrosine-protein phosphatase nonreceptor type 2). The full proteomics data set is available in the [Supplemental Data](#). We focused on the top 2 proteins associated with Δ LVEF based on adjusted P values, PIGR ($R=-0.74$; $P=0.0022$) and CPVL ($R=-0.74$; $P=0.0022$), where we observed that higher levels of PIGR and CPVL at baseline correlated with a more pronounced decline in LVEF after anthracyclines, but did not correlate with

Table 1. Characteristics of Patients Enrolled in the Discovery and Validation Cohorts

	Discovery cohort (n=34)	Validation cohort (n=32)
Demographic		
Age, y	60±14.7	64±14.4
Male	18	15
Female	16	17
White	27	26
Black	6	5
Asian	1	0
Other	0	1
Malignancy		
DLBCL	23	22
Hodgkin lymphoma	5	2
Follicular lymphoma	2	2
AML	1	2
Non-Hodgkin lymphoma	1	2
T-cell, histiocyte rich LBCL	1	2
Aggressive B-cell lymphoma	1	0
Cardiovascular treatments		
β-Blocker	2	8
ACE inhibitor	4	5
ARB	2	3
Cardiovascular risk factors		
Hypertension	12	14
Diabetes	3	4
Hyperlipidemia	10	10
Smoking	10	12
BMI	26.5±5.5	30.0±7.2
eGFR	93.3±18.8	83.0±23.7
Average doxorubicin dose	276.9±57.4	241.4±72.2
Systolic BP	132.5±16.2	128.1±15.6
Diastolic BP	77.7±8.6	77.5±8.6
Baseline LVEF	62.1±5.6	62.5±5.1
Relative decline in LVEF	−0.11±0.14	−0.05±0.13
Average time to follow-up echo, d	216.8±68.5	199.2±46.6

Data are represented as either mean ± SD or number positive. ACE indicates angiotensin-converting enzyme; AML, acute myeloid leukemia; ARB, angiotensin receptor blocker; BMI, body mass index; BP, blood pressure; DLBCL, diffuse large B-cell lymphoma; eGFR, estimated glomerular filtration rate; LBCL, large B-cell lymphoma; and LVEF, left ventricular ejection fraction.

baseline LVEF (Table 2; Figure S1C through S1E). Both associations remained significant after adjusting for clinical covariates, including age, sex, body mass index, and estimated glomerular filtration rate (Table 3). Importantly, we also found that relationships with both PIGR and CPVL remained significant when adjusting for baseline LVEF alone, as well as when adjusting for baseline LVEF in combination with age, sex, body mass index, and estimated glomerular filtration rate (Table 3). Given that the patient who experienced HF was an outlier in both

analyses, we also excluded this patient and found that the associations between baseline PIGR and CPVL persisted (Figure S2).

PIGR-IgA Axis Is Associated With Cardiotoxicity in Anthracycline-Treated Patients

We first focused on PIGR, which mediates the secretion of mucosal immunoglobulin across endothelial and epithelial surfaces (Figure 1A). We confirmed the relationship between baseline PIGR levels and Δ LVEF in the same discovery cohort using a commercially available ELISA (Figure 1B). We then assessed the relationship between pretreatment PIGR levels and Δ LVEF in a second prospectively enrolled cohort (n=32; Table 1). We observed a similar correlation between PIGR and Δ LVEF following anthracycline treatment (Figure 1C). As PIGR is involved in antibody secretion, we next assessed circulating immunoglobulin levels. Using a commercially available ELISA, we found that circulating IgA (immunoglobulin A) correlated with Δ LVEF in both cohorts (Figure 1D through 1E) and noted that PIGR and IgA were robustly correlated with each other (Figure 1F). Taken together, these data suggest that PIGR-mediated secretion of IgA correlates with the development of anthracycline cardiotoxicity.

PIGR-IgA Axis Is Associated With Cardiotoxicity in Anthracycline-Treated Mice

To facilitate subsequent mechanistic studies of PIGR and CPVL in anthracycline cardiotoxicity, we next asked if mice treated with doxorubicin exhibited similar relationships between these proteins and cardiac function as seen in patients. We treated wild-type mice with doxorubicin 5 mg/kg i.p. or PBS weekly for 4 or 8 weeks and observed a progressive decline in fractional shortening and increased dilation of the left ventricle (Figure 2A through 2C). At the time of euthanasia, plasma PIGR levels were increased after 8 weeks of doxorubicin treatment compared with control mice treated with PBS. In mice treated with doxorubicin, plasma PIGR levels correlated with fractional shortening ($R=-0.52$; $P=0.026$) and left ventricular diameter ($R=0.63$; $P=0.005$; Figure 2E and 2F). As plasma IgA correlated with Δ LVEF in patients treated with anthracyclines, we then measured circulating B cells and, specifically, IgA⁺ B cells in these mice using flow cytometry (gating strategy shown in Figure S3). Total circulating CD19⁺ B-cells transiently decreased after 4 weeks of doxorubicin treatment and then returned to baseline after 8 weeks (Figure 2G). However, circulating CD19⁺IgA⁺ B cells increased after 8 weeks of doxorubicin (Figure 2H), and the number of CD19⁺IgA⁺ B cells correlated with fractional shortening across all groups of mice (Figure 2I). Notably, we did not observe changes in CD19⁺ B cells or CD19⁺IgA⁺ B cells at either time point in cardiac tissue, suggesting that

Table 2. Human Proteomics Results

Protein	Pearson <i>R</i>	FDR <i>q</i> value
Polymeric immunoglobulin receptor (PIGR)	−0.74	2.20×10 ^{−3}
Probable serine carboxypeptidase CPVL (CPVL)	−0.74	2.20×10 ^{−3}
IQ domain-containing protein F3 (IQCF3)	−0.7	8.70×10 ^{−3}
Sesquipedalian-2 (SESQ2)	−0.69	8.70×10 ^{−3}
Versican core protein (CSPG2)	−0.69	8.70×10 ^{−3}
SHC-transforming protein 2 (SHC2)	−0.68	8.90×10 ^{−3}
Cryptic protein (CFC1)	−0.68	8.90×10 ^{−3}
Tyrosine-protein phosphatase nonreceptor type 2 (TCPTP)	−0.69	8.90×10 ^{−3}
Fibroblast growth factor 16 (FGF-16)	−0.68	9.80×10 ^{−3}
Aurora kinase B (AURKB)	−0.66	1.40×10 ^{−2}
Serpin A12 (Vaspin)	−0.66	1.40×10 ^{−2}
Fibroblast growth factor 18 (FGF-18)	−0.65	1.50×10 ^{−2}
Carbohydrate sulfotransferase 3 (CHST3)	−0.66	1.50×10 ^{−2}
Nectin-1, isoform gamma: Cytoplasmic domain (nectin-1 gamma:CD)	−0.65	1.50×10 ^{−2}
Zymogen granule membrane protein 16 (ZG16)	−0.65	1.80×10 ^{−2}
Ephrin type-A receptor 10 (EPHAA)	−0.64	2.10×10 ^{−2}
Alkaline phosphatase, placental type (Alkaline phosphatase, placental)	−0.64	2.10×10 ^{−2}
Serine/threonine-protein kinase 4 (STK4)	−0.63	2.60×10 ^{−2}
Disintegrin and metalloproteinase domain-containing protein 12 (ADAM12)	−0.63	2.60×10 ^{−2}
Lupus La protein: HTH La-type RNA-binding domain (LA:HTH)	−0.62	3.50×10 ^{−2}
LETM1 domain-containing protein 1 (HCCR1)	−0.61	3.70×10 ^{−2}
Cytochrome C oxidase subunit 6C (COX6C)	−0.61	3.70×10 ^{−2}
Plexin domain-containing protein 1 (PXDC1)	0.62	3.70×10 ^{−2}
Carbonic anhydrase7 (Carbonic anhydrase VII)	−0.61	3.90×10 ^{−2}

Shown are proteins identified from the SomaScan 7000 platform that met significance for correlation with change in LVEF (Pearson *R* value) based on a FDR <0.05. FDR indicates false discovery rate; and LVEF, left ventricular ejection fraction

the CD19⁺IgA⁺ B-cell response is systemic rather than cardiac-specific in mice treated with doxorubicin (Figure S3B and S3C). Together, these data implicate the PIGR-immunoglobulin axis as a systemic response in patients and in rodent models of anthracycline cardiotoxicity.

Circulating CPVL Is Associated With Cardiotoxicity in Anthracycline-Treated Patients and Mice

We next sought to validate the observation that CPVL correlates with ΔLVEF, as observed by aptamer-based

proteomics in our discovery cohort (Figure 3A). As no commercially available ELISA kits or published quantitative methods were available for human CPVL, we performed immunoblotting on patient plasma, comparing baseline plasma CPVL levels in patients who had minimal change in LVEF with those who had a >20% decline in LVEF following anthracyclines. We observed higher CPVL levels in patients with a >20% decline in LVEF (Figure 3B and 3D). We then performed a similar analysis in samples from our validation cohort and again found higher circulating CPVL in patients with decreased LVEF (Figure 3C and 3E).

Table 3. Adjustment of Top Human Proteomics Findings for Clinical Factors

	N	Adjusted for age, sex, BMI, eGFR		Adjusted for baseline LVEF		Adjusted for all factors	
		Coefficient (95% CI)	<i>P</i> value	Coefficient (95% CI)	<i>P</i> value	Coefficient (95% CI)	<i>P</i> value
CPVL (Proteomics)	34	−0.082 (−0.112 to −0.053)	<0.0001	−0.084 (−0.112 to −0.055)	<0.0001	−0.079 (−0.109 to −0.049)	<0.0001
PIGR (Proteomics)	34	−0.154 (−0.206 to −0.102)	<0.0001	−0.151 (−0.205 to −0.097)	<0.0001	−0.150 (−0.206 to −0.095)	<0.0001
PIGR (ELISA)	66	−0.050 (−0.078 to −0.022)	0.0006	−0.047 (−0.072 to −0.022)	0.0003	−0.049 (−0.075 to −0.023)	0.0004

Adjustment for age, sex, BMI, eGFR and baseline LVEF alone or in combination was performed using ordinary least squares linear regression for the association between CPVL and PIGR and change in LVEF. Analyses were performed in both the discovery cohort proteomics data set as well as pooled ELISA data for PIGR discovery and validation cohorts. Shown are *P* values for the CPVL or PIGR coefficients. BMI indicates body mass index; CKD-EPI, chronic kidney disease epidemiology collaboration; CPVL, carboxypeptidase vitellogenic-like protein; eGFR, estimated glomerular filtration rate; LVEF, left ventricular ejection fraction; and PIGR, poly immunoglobulin receptor.

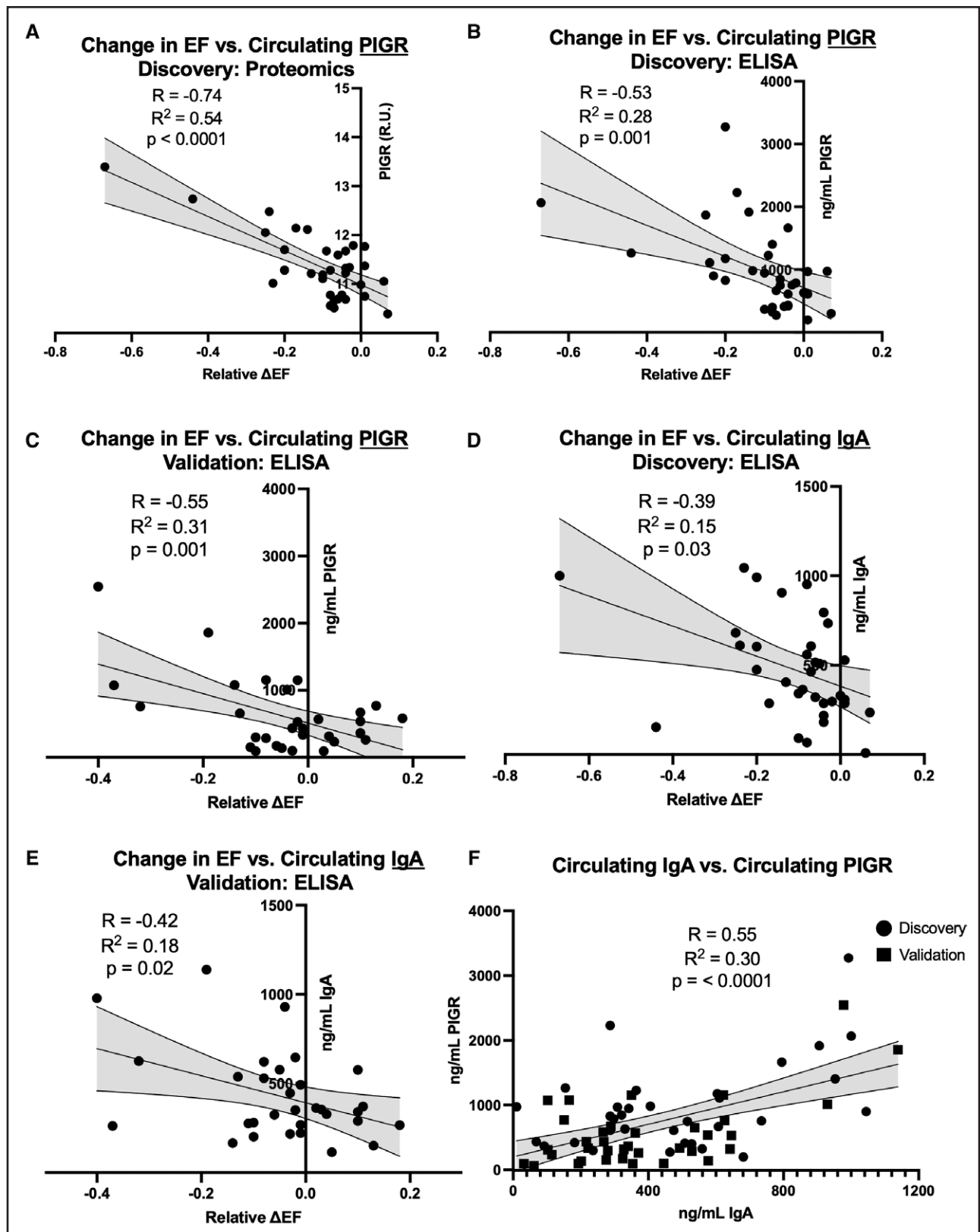


Figure 1. Elevated levels of circulating PIGR (poly immunoglobulin receptor) and IgA (immunoglobulin A) are associated with cardiac dysfunction after anthracycline treatment in patients.

A, Baseline plasma PIGR from SomaScan 7000 plotted against relative change in left ventricular ejection fraction (LVEF) for discovery cohort patients (N=34). **B**, Baseline plasma PIGR was measured by ELISA and plotted against relative change in LVEF for discovery cohort patients. **C**, Baseline plasma PIGR was measured by ELISA and plotted against relative change in LVEF for validation cohort patients (n=28). (Continued)

Macrophages have been reported to express CPVL, whereas this protein is not known to be expressed in CD8⁺ T cells.¹⁵ To begin to assess CPVL in mouse models, we compared lysates from mouse bone marrow-derived macrophages with lysates from CD8⁺ T cells. As expected, we observed expression of CPVL in macrophage but not T-cell lysates (Figure S4). We next asked if doxorubicin treatment induces CPVL expression or secretion in macrophages as the mechanism for increased circulating CPVL. We generated human cell line-derived macrophages from the THP1 monocytic cell line and treated them with increasing doses of doxorubicin. We observed no change in CPVL expression in the lysates, and CPVL was undetectable in the supernatant, suggesting that doxorubicin does not induce CPVL expression or secretion (Figure S5A). We then compared plasma CPVL levels in mice treated with saline or 8 weeks of doxorubicin by immunoblot and observed an increase in plasma CPVL (Figure 3F and 3G). Circulating levels of CPVL correlated with fractional shortening in saline and 8-week doxorubicin-treated mice (Figure 3H). Together, these data indicate that doxorubicin induces circulating CPVL, and that this effect correlates with the development of systolic dysfunction.

Circulating PIGR and CPVL Are Associated With Survival in Anthracycline-Treated Patients

Finally, we asked if the protein biomarkers identified in our patient cohorts were also associated with oncological outcomes and overall survival. We found that both baseline PIGR ($R=0.35$; $P=0.0061$) and IgA ($R=0.34$; $P=0.0087$), but not CPVL ($R=0.04$; $P=0.84$), correlated with levels of LDH, an important prognostic marker in lymphoma (Figure 4A; and Figure S6A and S6B). Neither baseline CPVL nor PIGR levels correlated with the doxorubicin dose administered (Figure S6C through S6E).

Within the combined cohort, 64 patients had at least 1 baseline biomarker level and survival outcomes data available. Of those, 6 died and 14 experienced cancer progression. The specific causes of death varied and included infection, stroke, and cancer relapse. Given the limited sample size and follow-up time, with few mortality events, only a limited exploratory analysis could be conducted to investigate the correlation of biomarkers with survival outcomes. Neither PIGR nor IgA levels differed by relapse status or binary survival status (Figure S6F through S6I). Cox proportional hazards modeling revealed that baseline PIGR, as a continuous variable, was associated with an increased hazard ratio for overall

survival, an effect that persisted after adjustment for age and international prognostic index score (hazard ratio, 5.63 [95% CI, 1.11–28.43]; $P=0.04$; Figure 4B). However, the same effect was not seen in the hazard ratio for progression-free survival (Figure 4C).

As an optimal cutoff for PIGR or IgA with regard to cardiac risk or survival has not yet been established, outcomes were compared by baseline biomarker quartile in an observational analysis. Mortality events were found to be clustered in the highest quartile of PIGR expression, with a difference between time to mortality in the highest quartile compared with any and all other quartiles, as assessed by Kaplan-Meier analysis ($P=0.034$; Figure 4D). In the discovery cohort, decreased overall survival time was seen in patients in the top quartile of baseline CPVL expression, compared with all other quartiles, by Kaplan-Meier analysis ($P=0.0004$; Figure 4E). Higher baseline CPVL levels were found in patients who did not survive by Wilcoxon signed-rank test ($P=0.0041$), but no differences in baseline CPVL were seen based on cancer progression status ($P=0.81$; Figure 4F; Figure S6J). Cox proportional hazards modeling using CPVL as a continuous variable did not reveal an increased hazard ratio for progression-free survival or overall survival (Figure 4G and 4H). Together, these data suggest that the immunologic markers CPVL and especially PIGR may be associated with worse overall survival in patients with cancer. Given the limited sample size and lack of pre-specified biomarker cutoffs, these observations must be taken as an exploratory analysis, as the relationship with cancer- or cardiac-specific mortality could not be ascertained in the present study. A summary of our findings is shown in Figure 5.

DISCUSSION

Here we used aptamer-based proteomics to uncover 2 circulating immunologic factors associated with anthracycline cardiotoxicity. These data highlight a role for systemic inflammation and associated immunologic biomarkers in anthracycline cardiotoxicity, while providing further proof of concept for using plasma proteomics as a tool to investigate mechanisms of anthracycline toxicity. Our study identified PIGR, which assists in mucosal antibody secretion, and CPVL, which is presumed to play a role in antigen presentation. Pretreatment levels of these plasma proteins were associated with cardiac dysfunction and overall mortality in patients treated with anthracyclines, including after adjustment for demographic and clinical factors. These findings were validated in a second cohort of patients as well as in a preclinical

Figure 1 Continued. **D**, Baseline plasma IgA was measured by ELISA and plotted against relative change in LVEF for discovery cohort patients. **E**, Baseline plasma IgA was measured by ELISA and plotted against relative change in LVEF for validation cohort patients. **F**, Baseline plasma PIGR vs plasma IgA for both cohorts. Statistical analysis by Pearson correlation; 95% CIs are shown, and exact nominal P values are shown. EF indicates ejection fraction.

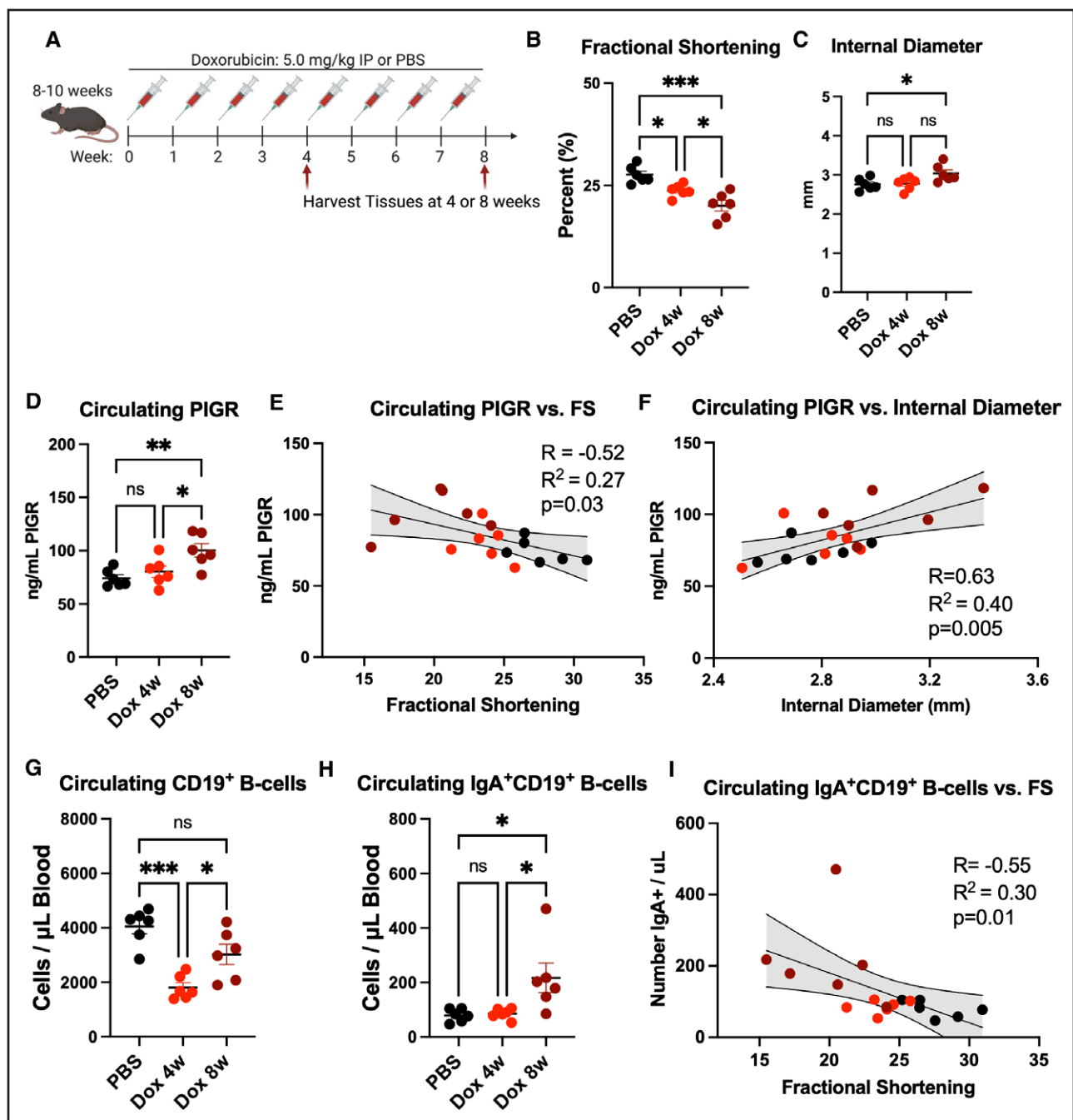


Figure 2. Circulating PIGR (poly immunoglobulin receptor) and IgA⁺ (immunoglobulin A) B cells are associated with doxorubicin-induced cardiac dysfunction in mice.

A, Schematic of in vivo doxorubicin administration. Wild-type (WT) mice aged 8 to 10 weeks were given 5.0 mg/kg doxorubicin weekly by intraperitoneal injection or an equivalent volume of sterile PBS. Tissues were harvested after 4 or 8 weeks ($n=6$ /group). **B** and **C**, Sedated echocardiography was used to determine fractional shortening (**B**) or internal ventricular diameter (**C**) 1 day before tissue harvest. **D**, PIGR was measured from plasma by ELISA for mice treated with saline or doxorubicin. **E** and **F**, Plasma PIGR plotted vs fractional shortening (**E**) or internal diameter (**F**). **G** and **H**, Flow cytometry was used on whole blood to measure CD19⁺ B-cells (**G**) or IgA⁺CD19⁺ B cells (**H**). **I**, IgA⁺ B cells vs fractional shortening for mice across all groups. Statistical analysis by 1-way ANOVA with Tukey multiple comparison test (**B** through **D**, **G**, and **H**) or Pearson correlation; 95% CIs are shown (**E**, **F**, and **I**; P values shown). * $P<0.05$, ** $P<0.01$, and *** $P<0.001$. ns indicates no significance. Created in BioRender. Bayer, A. (2026) <https://BioRender.com/y70ssi8>

mouse model of doxorubicin cardiotoxicity. This suggests that preexisting differences in both the innate (CPVL) and adaptive (PIGR) immune responses may predispose patients to cardiotoxicity. Ultimately, measurement

of these biomarkers in patients before treatment could assist with categorizing patients as high risk or low risk to guide the frequency of echocardiography, the utilization of cardioprotective medications, and referral to a

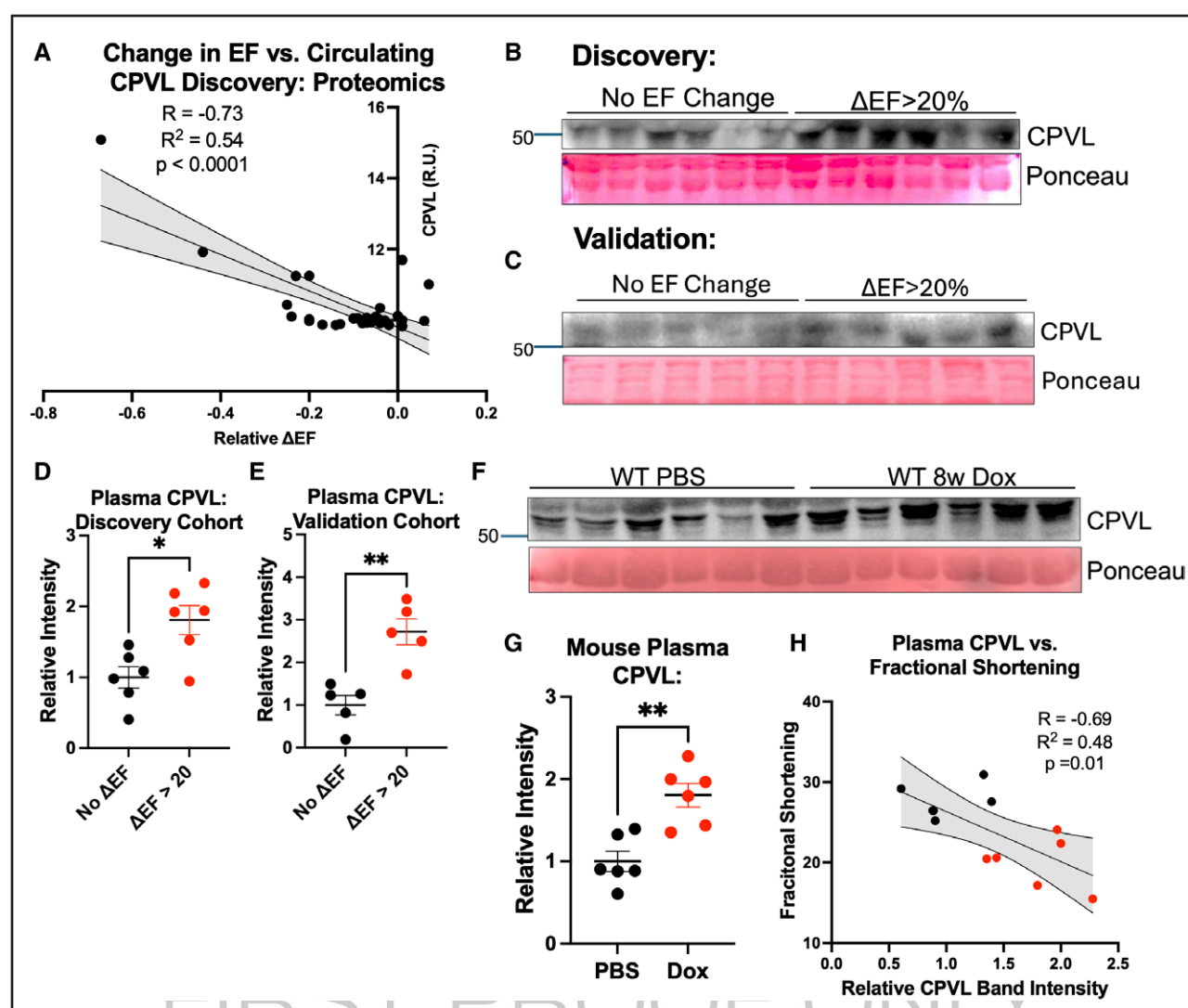


Figure 3. Circulating CPVL (carboxypeptidase vitellogenic-like protein) is associated with cardiac dysfunction after anthracycline treatment.

A, Raw data for baseline plasma CPVL from SomaScan 7000 plotted against relative change in left ventricular ejection fraction (LVEF) for discovery cohort patients (N=34). **B**, Western blot was used to measure baseline plasma CPVL in selected patients from the discovery cohort (n=6) who had no change in LVEF and was compared with CPVL levels from patients with a drop in LVEF >20% (n=6). **C**, Western blot was used to measure baseline plasma CPVL in selected patients (n=5) from the validation cohort with no change in LVEF compared with n=5 with a drop in LVEF >20%. **D** and **E**, Quantification of Western blot results from **B** and **C**. **F**, Western blot was used to measure plasma CPVL in mice treated with PBS or doxorubicin for 8 weeks (n=6/group). **G**, Quantification of Western blot results from **F**. **H**, Plasma CPVL band intensity vs fractional shortening from mice treated with PBS or doxorubicin for 8 weeks. All Western blots were analyzed by ImageJ. Statistical analysis by *t* test (**D**, **E**, and **G**) or Pearson correlation (**A** and **H**); 95% CIs are shown (*P* values shown). **P*<0.05 and ***P*<0.01. ns indicates no significance. EF indicates ejection fraction; and WT, wild-type.

dedicated cardio-oncology program. Although patient-to-patient genetic variability that contributes to variable CPVL and PIGR levels in people cannot be recapitulated using mice of the same background, both CPVL and PIGR correlated with LV function in doxorubicin-treated mice compared with saline-treated mice. There is increasing acknowledgment that genetic factors contribute to anthracycline cardiotoxicity, although genetic predictors are not yet robust enough to be incorporated routinely into clinical risk assessment.¹⁶ Genetic variants in PIGR have been shown to correlate with nasopharyngeal

cancer, and circulating PIGR levels have also been linked to other types of cardiovascular disease.^{17–19}

This study highlights the potential for high-throughput proteomics assays to identify new circulating biomarkers of anthracycline cardiotoxicity. A similar approach was used in patients treated with the cardiotoxic proteasome inhibitor carfilzomib for multiple myeloma.²⁰ This study used Olink proteomics to measure a panel of 92 cardiac-specific biomarkers in 25 patients treated with carfilzomib, 6 of whom developed a clinically significant decline in LVEF. To our knowledge, the full panel of

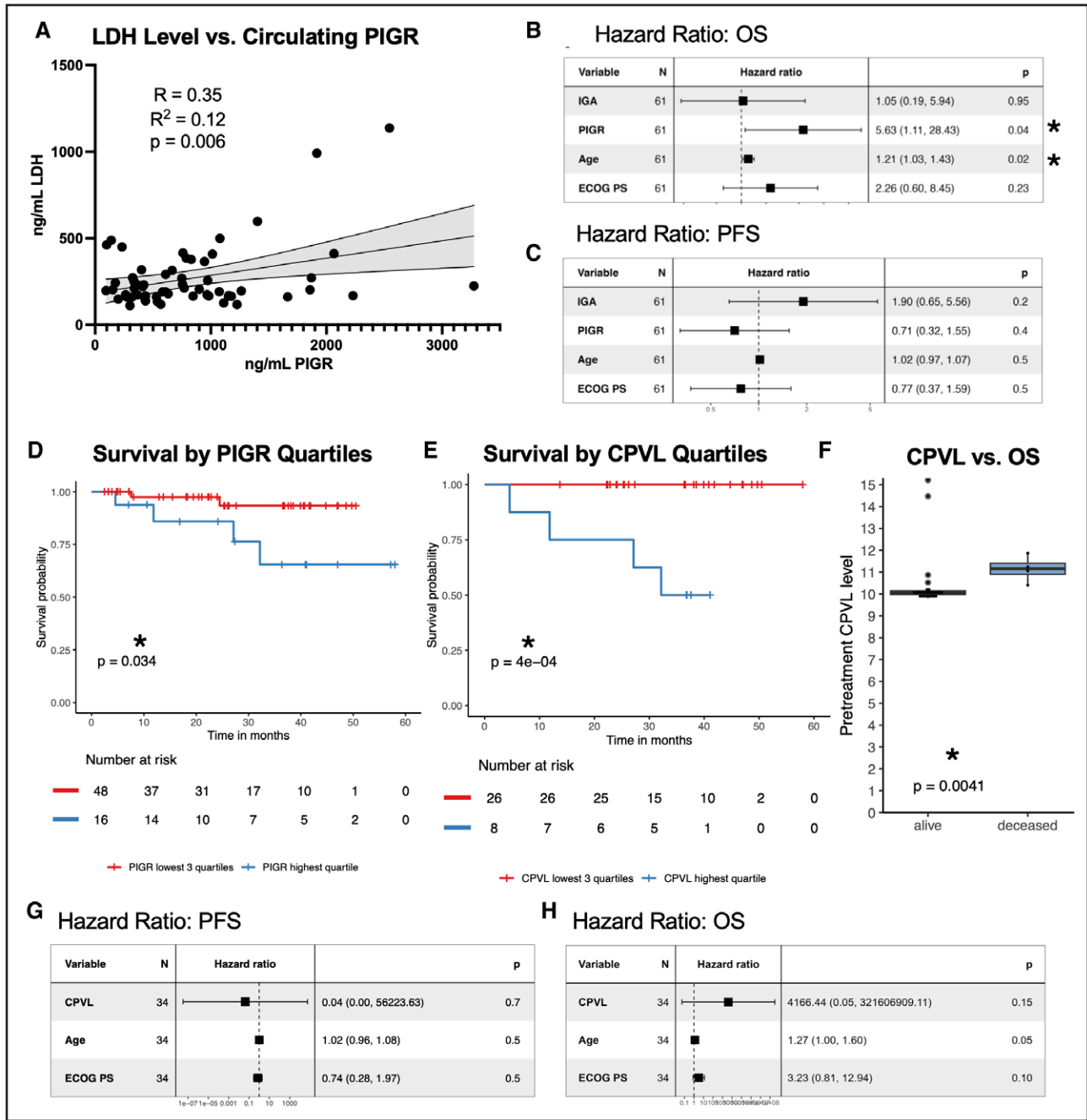


Figure 4. Circulating PIGR (poly immunoglobulin receptor) and CPVL (carboxypeptidase vitellogenic-like protein) are associated with cancer outcomes. **A**, Correlation of circulating LDH with PIGR by ELISA from combined discovery and validation cohorts (n=62). **B** and **C**, Cox proportional hazards analysis for hazard ratios for IgA and PIGR in overall survival (**B**) or progression-free survival (**C**) adjusted for age and ECOG score. **D** and **E**, Kaplan-Meier survival curves in patients with the highest quartile of PIGR (**D**) or CPVL (**E**) compared with the bottom 3 quartiles. **F**, Boxplot with CPVL levels split by overall survival status. **G** and **H**, Cox proportional hazards analysis for hazard ratios for CPVL in progression-free survival (**G**) and overall survival (**H**) adjusted for age and ECOG score. Statistical analysis by Pearson correlation (**A**), Cox proportional hazards model (**B**, **C**, **G**, and **H**), Kaplan-Meier (**D** and **E**), or Wilcoxon signed-rank test (**F**); exact *P* values shown.

≈5000 proteins measured by Olink has not been previously assessed in the context of cardiovascular toxicity from anthracyclines or other anticancer agents. Inflammation has recently emerged as a central mediator of anthracycline cardiotoxicity, having long been recognized to play a broad role in cardiovascular disease.^{21–23} Our group recently showed that cytotoxic T cells contribute to cardiotoxicity in mice treated with doxorubicin,⁵ implicating the adaptive immune response as a driver of chronic inflammation. Recent work has also shown that B-cell depletion is protective in doxorubicin-treated mice.¹² Our observations on the role

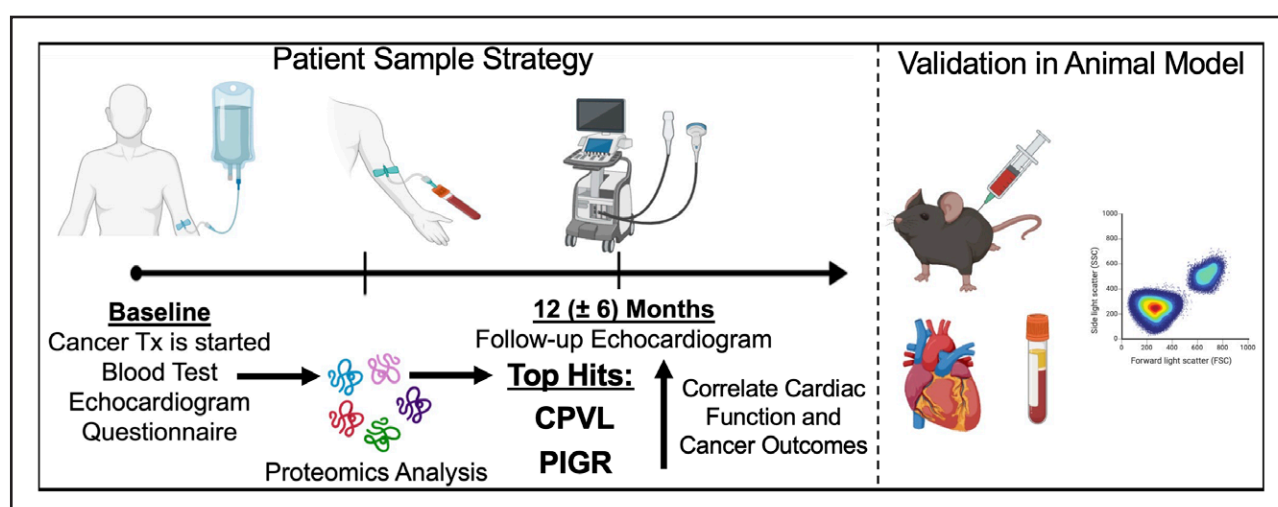


Figure 5. Overview of strategy and findings.

CPVL indicates carboxypeptidase vitellogenic-like protein; PIGR, poly immunoglobulin receptor; and Tx, treatment. Created in BioRender. Bayer, A. (2026) <https://BioRender.com/y70ssi8>

of the PIGR/IgA⁺ B-cell axis in doxorubicin cardiotoxicity build upon this prior work and define a more specific element of the humoral immune response. It remains unknown whether IgA is causative in the disease and whether it is deposited in the heart. Cardiac antibody deposition, including IgA, has been described as a component of other etiologies of HF.^{24–27} The inciting antigen for both B- and T-cell induction in doxorubicin cardiotoxicity remains unknown, and, if shared, could represent a novel cardioprotective target. More broadly, the immune response in anthracycline toxicity shares features with that of immune checkpoint inhibitor myocarditis. Both toxicities involve antigen specific cytotoxic CD8⁺ T-cell infiltration and myeloid cell-sourced CXCL9/10.^{28,29} As such, it is possible that inflammatory biomarkers such as PIGR and CPVL have utility in immune checkpoint inhibitor cardiotoxicity as well.

These findings also highlight the role of macrophages, which both amplify pathogenic inflammation and contribute to reparative functions in doxorubicin cardiotoxicity.^{11,30} CPVL is primarily expressed in monocytes and macrophages. It is hypothesized to play a role in antigen presentation and macrophage differentiation; however, a definitive function has not yet been described.^{15,31} It is plausible that CPVL contributes to antigen presentation leading to B- and T-cell activation, but it is not thought to be secreted,¹⁵ as supported by our observations in THP1-derived macrophages. Additional mechanistic work is needed to understand how CPVL is released into human plasma. Our previous work showed increased circulating monocytes/macrophages in doxorubicin-treated mice,⁵ 1 potential contributor to increased circulating CPVL. However, CPVL is highly expressed in the heart, brain, and kidney¹⁵ and may play a different functional role depending on cell type or tissue, including that of other leukocytes.

Future studies will further delineate functional roles for CPVL in the response to anthracyclines in immune and nonimmune cells.

The current study focused on patients with hematologic malignancies, which are often accompanied by a robust inflammatory milieu. Whether a specific type or stage of malignancy is associated with specific blood biomarkers or mechanisms of cardiotoxicity remains understudied. In our prior work,⁶ baseline levels of PIGR were not associated with early cardiac dysfunction in patients with breast cancer. CPVL was not assessed in this prior cohort, as it was not included in the earlier Somascan platform. Conversely, we had previously reported the heme-binding, macrophage-associated protein hemopexin as associated with anthracycline cardiotoxicity in patients with breast cancer,⁶ but in the current study, hemopexin did not correlate with change in LVEF (data not shown). These observations suggest that tumor type may influence cardiotoxicity manifestations in patients. We further acknowledge that while we focused on the inflammatory proteins CPVL and PIGR, other findings from our proteomics data set will be further explored in future studies, using additional unbiased methodologies to broadly probe the plasma proteome. Similar approaches have been taken by others in broader contexts, highlighting the power of probing the circulating proteome.^{20,32–35} These studies did not identify PIGR or CPVL, likely due to one of several reasons. Several studies are not specific to anthracycline cardiotoxicity,^{20,33,35} use patients of different ages,³² or define cardiotoxicity more broadly than decreased LVEF.³⁴ As noted, the type of malignancy might dictate the inflammatory marker profile. However, several of these studies identified other inflammatory markers as correlating with cardiovascular outcomes, in line with our findings and a major goal of our study.

There were few mortality events due to the size of the cohort and duration of this study. When present, mortality events clustered in the highest quartile of baseline PIGR and CPVL expression, and some measures of overall survival correlated with pretreatment biomarkers. The causes of death were varied, and no single cause of mortality was identified to be associated with baseline biomarker status. Interestingly, CPVL^{36–38} and PIGR^{39–41} have been associated with both pro- and antitumor effects in several malignancies in patients and preclinical studies. PIGR is also expressed on B-cell lymphoma cells themselves,⁴² raising the possibility that systemic immune markers may represent both cancer severity and cardiotoxicity. Larger studies over longer time intervals will be required to elucidate the specific causes of mortality associated with these inflammatory biomarkers. Future functional studies will allow for the identification of specific risks associated with baseline inflammatory biomarkers, as well as any trade-offs that exist between cardioprotection and oncological outcomes. While this study introduces novel immunologic factors correlating with anthracycline cardiotoxicity, we acknowledge limitations to the approach. First, given the cost of performing aptamer-based proteomics in the full cohort, we elected to perform this technique in a smaller discovery cohort and then validate key findings with more targeted methods, such as ELISA. It is noteworthy that the 2 cohorts, although drawn from the same institution, were enrolled during different time periods and cared for by different clinical teams, with natural variations in patient management, sample handling, and operational practices across enrollment periods. Nonetheless, external validation will be an important future step in applying our findings to the clinical setting. In addition, while aptamer-based proteomics has been widely used to identify circulating biomarkers in several disease states, not all aptamers have published data to support their specificity, such as protein quantitative trait loci or confirmation by complementary methods such as mass spectrometry. Here, we used alternative techniques to demonstrate similar changes in these proteins in mice and humans treated with doxorubicin. Because each orthogonal assay requires optimization, technical replication, and substantial experimental resources, it was not feasible to pursue all 24 candidate proteins at this level of depth.

The functional contributions of PIGR and CPVL to cardiotoxicity remain unknown, justifying future mechanistic studies of associated pathways. PIGR and CPVL may provide important mechanistic information about doxorubicin cardiotoxicity, but the cohort sizes in this study preclude a rigorous assessment of sensitivity, specificity, and predictive value. Given the small size of the discovery cohort, we selected a straightforward statistical approach for the current study. Other single-protein selection methodologies such as LASSO, or unsupervised clustering methods to highlight top hub proteins in network analyses, could

be considered as these findings are extended to larger external validation cohorts. Thus far, we have validated PIGR and CPVL in a small, internal cohort. However, it will be essential to evaluate these circulating biomarkers in larger cohorts and in patients with different malignancies. The current study population consists of middle-aged to older adults. Therefore, the generalizability of these findings to pediatric patients, who frequently receive anthracyclines, remains to be determined. While we found no overt correlation between PIGR or CPVL and cancer progression, it is possible that our analyses did not account for confounding oncological factors. In addition, this study only utilized samples drawn pretreatment and 3 months post-treatment; future studies will aim to perform a more thorough characterization of CPVL and PIGR over time. While some error may have been introduced by a few echocardiograms being read by clinical visual interpretation only, and there was some variability in the time to our follow-up echocardiography analysis, a majority of the echos occurred in the 5- to 7-month range, consistent with prior data indicating that 95% of cardiotoxicity can be detected within the first year.³ We also acknowledge that the decrease in LVEF is modest, and the clinical significance of the association between decreased LVEF and PIGR or CPVL levels requires further study. Finally, while our findings are comparable between humans and mice, in humans we focused on baseline plasma evaluation, whereas in mice we compared control with doxorubicin-treated mice at the terminal time point. In addition the use of tumor-bearing mice may recapitulate human findings more accurately. Finally, PIGR and CPVL were not associated with progression-free survival, indicating that these biomarkers are unlikely to reflect tumor progression directly. The observed association with overall survival may reflect broader treatment-related or systemic factors, such as immune dysregulation or noncancer-related morbidity and mortality. The number of survival events was limited, with considerable competing risks in this population. As a result, survival analyses in this small cohort should be interpreted cautiously and considered exploratory and hypothesis-generating.

CONCLUSIONS

Taken together, these data uncover a role for the plasma proteins PIGR and CPVL in anthracycline cardiotoxicity in patients with hematologic malignancies. These proteins represent potential biomarkers and may functionally contribute to cardiotoxicity. Our results highlight the need for additional investigation to characterize inflammatory markers and immunologic contributors to cardiotoxicity in patients treated with doxorubicin.

ARTICLE INFORMATION

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Disclosures

Dr Asnani is a cofounder and serves on the Board of Directors for Corventum, Inc, and receives patent royalties from Mass General Brigham. The other authors report no conflicts.

Supplemental Material

Supplemental Methods
Figures S1–S6
Full Proteomics Data Set

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